

Bolo Bill — PoC baseline and evidence register

What exists, what has been proven, and what has not. Everything downstream cites this page. Written to be handed to a diligence process unedited.

1. What the PoC actually does

A voice-first billing web app running on a shopkeeper's own Android phone. The shopkeeper speaks; the bill assembles line by line; a UPI QR carrying the exact amount is shown; a numbered receipt is produced.

Capability	State	Detail
Voice → line items	Working	Multi-item in one breath. "2kg potato, 1kg tomato, ½kg onion" → three priced lines.
Catalog & price entry by voice	Working	Admin mode; owner-only; confirmation gate on ambiguous rate-vs-pack.
Confidence gating	Working	accept ≥0.72 / confirm ≥0.42 / reject. Deliberately pessimistic.
Spoken aliases per SKU	Working	Shop teaches its own words: "aloo" → its potato. The compounding asset.
UPI collection	Working	Deep link + QR, exact amount. We never touch funds.
Cash settlement	Working	Recorded on the shopkeeper's word — first-hand and honest.
Customer identity & repeat orders	Working	Spoken phone number recalls last 5 baskets; one tap reloads one at today's prices.
GST-aware receipt	Working	Gapless serial per shop per financial year, allotted by the database.
58mm thermal print	Partial	32-column render + Web Bluetooth ESC/POS. Never tested on real hardware.
WhatsApp receipt	Working	Free wa.me hand-off. Business API endpoint built but uncredentialed.
Hands-free wake word	Unstable	"Hey Chitti". Works, then stalls. Under active repair.
Inventory deduction	Not built	Schema exists; no logic.
Offline billing	Not built	Requires the network today.
Payment reconciliation	Not built	Deliberate — see risk R3.

2. Stack as built

ASR: Sarvam saarika:v2.5, cloud, per-utterance, per-shop language code.

Understanding: a deterministic grammar parser. **No LLM anywhere in the utterance path.** Chosen because a grammar's failures are legible — when it cannot parse, it knows, and that feeds the confidence gate. A model that always returns something confident is the exact failure mode this product cannot afford.

Backend: Python/FastAPI on Vercel serverless · Supabase Postgres over PostgREST.

Client: vanilla-JS PWA, Material 3 tokens. Chrome Web Speech API for wake-word spotting; Web Bluetooth for ESC/POS.

Languages: Tamil, English, Hindi, Malayalam, Telugu, Kannada — code-mixed, per shop. Numerals, units, aliases and phonetic rules are data files, not code.

3. Validated — with the evidence

Claim	Evidence
The parser is correct across six languages	1,561 automated checks , including a 1,003-combination matrix of every wake word, command alias and mode alias against a catalog stocked with deliberately colliding names (bill paper, cashew, chutney). Zero collisions.
It survives real ASR output, not clean text	Tested by generating Tamil/Hindi/Kannada speech and passing it through the live Sarvam endpoint. This found four silent mispricing classes that clean-text tests missed.
Multi-item dictation works	4 items, mixed units, one breath, Tamil → 4 correct priced lines.
Receipt numbering is gapless under load	8 simultaneous finalises → 8 distinct sequential numbers, no duplicates, no gaps.
A real shop used it, and moved up the interaction ladder	Ragas Coffee ran ~50 bills in a week and progressed from press-to-talk to the wake word within that week . Small numbers, but the direction is the signal: given a hands-free option, the operator took it. That is the product thesis surviving one week of contact with a real counter.
Six languages are genuinely configured	Numeral tables filled for all six; ~170 aliases added after discovering the non-Tamil packs were Tamil tables with a few native words sprinkled in.

The silent-error work is the strongest technical signal. Three real defects found and fixed: ₹115 of goods billing as ₹12.50 (multi-item collapse); 500g of a ₹50/kg item billing ₹25,000 (unit mismatch); 300g of onion billing ₹0.07 (unrecognised numeral). Each was silent. Each is now a regression test. This is the discipline the product's core promise depends on.

4. NOT validated — the honest register

These go in the deck's risk slide, not buried. A diligence process finds them either way.

Gap	Status
Real-shop volume	Ragas Coffee: ~50 bills over one week , a genuine café with a 24-SKU catalog. Enough to prove the loop closes in a real shop; too few and too short to claim retention, or accuracy at a statistically useful confidence. VALIDATE whether staff or owner operated it, and capture error rate — the app is not yet instrumented to report either.
Unit cost — the central one	Target is ₹10/shop/month with ₹0 cloud inference . The PoC calls a paid cloud ASR on every utterance. At ~24,000 utterances/shop/month this is the constraint the whole business model turns on, and it is not met today . VALIDATE Sarvam per-second rate.
Latency	Target p95 speech-end → line on screen is 1,150 ms. Measured ASR round trip is ~2,200 ms plus parse. Not met. Cloud round trip is the cause; on-device ASR is the answer.
Noisy environment	Only tested against a podcast played deliberately (background –45 dB vs voice –13 dB). Not tested in a real shop at 70–80 dB(A) with customers.
Real accents	All synthetic-speech testing used macOS TTS voices. No real human speaker other than the founder has been transcribed.
Offline	Not built. CLAUDE.md calls offline non-negotiable; the PoC requires the network.
Thermal printer	Never connected to a physical printer. Cheap ESC/POS units are Bluetooth Classic, which a browser cannot open at all — needs the Android app.
Wake word stability	Stalls after first use; three causes found and fixed, unverified on a handset.
Staff adoption	No non-founder has been trained on it or observed using it under time pressure.

5. What this baseline means for the raise

This is a working prototype with one week of real use at one design partner, not a pilot with traction. Positioned honestly it is still a strong pre-seed artifact: six languages, a tested grammar, a real receipt series and a live URL, built solo. Positioned as traction it will not survive a technical diligence call.

The two questions a competent investor asks first are the two in the register above: *what does an utterance cost you*, and *does it work in a loud shop with someone else's voice*. The roadmap must answer both, and the funding ask should be visibly sized to do so.